

Adult masking diagnostic

This one-page report measures the precise limitation targeted by Phase 3: invalid or unrealistic coalition states after preprocessing.

Adult Income is a strong next dataset because it has many categorical groups. Zero-masking creates impossible hidden one-hot states, while empirical_background preserves valid hidden groups by copying from real transformed training rows.

The diagnostic shows hidden categorical validity rising from 0.0000 under zero_mask to 1.0000 under empirical_background. It also reduces exact-zero hidden numeric states and changes nearest-train distance from 2.1893 to 1.3329.

This does not by itself prove better full-model explanation fidelity, but it does prove that the masking improvement directly fixes the coalition-construction problem more clearly on Adult Income than on Covertype.

